

Rahul Madhav Deshpande

(510)-953-2367 | Baltimore, MD | rahulmadhavd04@gmail.com
<http://www.linkedin.com/in/rahulmadhavdeshpande> | rahulmadhavdeshpande.com

RESEARCH INTERESTS

Image-guided robotics and visual servoing for minimally invasive surgery and interventional procedures, grounded in a background in photoacoustic transducer design and optical/acoustic simulation, with the aim of bridging academic research and clinical device translation.

EDUCATION

Johns Hopkins University — Whiting School of Engineering

Baltimore, MD

M.S.E., Mechanical Engineering — Robotics and Intelligent Systems track

Aug 2026 – Present

Pennsylvania State University, University Park

State College, PA

Major: BS Mechanical Engineering | Minors: Mechatronics

Graduation Date: May 2026

RESEARCH EXPERIENCE

Biophotonics and Ultrasound Imaging Lab

State College, PA

Undergraduate Research Assistant — Advisor: Dr. Sri-Rajasekhar Kothapalli

Aug 2025 – Present

- Spearheaded the development of focused Transparent Ultrasound Transducer (f-TUT) arrays, transitioning fabrication workflows from planar to focused geometries to enable high-resolution imaging while optimizing device yield
- Designed the flex PCB and interconnect for a 70-element array, routing that many independent channels through a package small enough to fit the probe without degrading signal integrity, along with a separate impedance-matching board for the same array
- Developed and trained neural networks for photoacoustic (PA) image enhancement using parallel PA and ultrasound images to improve resolution and reduce noise
- Engineered and validated digital phantoms with NIRfast and K-wave simulation toolsets to generate synthetic training datasets, improving neural model generalization for tissue-mimicking conditions
- Conducted piezoelectric material analysis across multiple cuts of Lithium Niobate and PMN-PT samples, using PiezoCAD simulations to derive electromechanical parameters that informed next-generation TUT design decisions

Department of Mechanical Engineering, Penn State

State College, PA

Undergraduate Research Assistant

Feb 2024 – Dec 2024

- Developed and implemented LiDAR-based object-avoidance algorithms using Python and ROS 2, enhancing real-time obstacle detection accuracy by 20% compared to baseline Euclidean clustering methods
- Led integration of ROS 2 with MATLAB/Simulink to streamline data transfer and control tuning, reducing development time by 30% and improving closed-loop stability during dynamic path tests
- Integrated Crazyflie drones with a Vicon motion-capture system and used its high-precision ground-truth data to calibrate onboard perception sensors, achieving ± 2 cm localization accuracy

PUBLICATIONS

Deshpande, R.M. “Prettier but Less Detectable: Deep-Network Reconstruction Suppresses Lesion Detectability in Limited-View Linear-Array Photoacoustic Tomography.” (In preparation, 2026)

Deshpande, R.M., Kumar, V., Dharmagadi, P. “Handwriting Analysis using Hybrid CNN-Transformer Framework for Adaptive Stress Detection via Biometric Stylus Sleeve.” (In preparation, 2026)

SELECTED PROFESSIONAL AND TEACHING EXPERIENCE

Siemens Centre of Excellence (GTTC)

Mysore, IND

Mechanical Engineering Intern

Jul 2025 – Aug 2025

- Programmed and executed CNC lathe and milling operations using Siemens Sinumerik (G/M codes) to machine high-tolerance prototype lunar rover parts, achieving dimensional accuracy within ± 0.02 mm and reducing machining cycle time by 15%
- Developed and validated robotic kinematics models in ROS 2, applying forward and inverse kinematics to optimize joint trajectory planning; improved industrial arm path repeatability by 22% through precise encoder calibration

NASA L'SPACE Mission Concept Academy

Remote, USA

Mission Power Subsystem Lead

Jan 2025 – Jun 2025

- Contributed to the design of a low-cost Aerobot System for Venus exploration under a \$200M budget and 50 kg mass limit, with emphasis on power subsystem design and environmental survivability
- Led design and trade studies for the Venus Aerobot Power Subsystem, selecting high-efficiency multi-junction solar panels and Li-ion storage to deliver ≥ 200 W continuous power under 150°C, 92-bar conditions
- Applied NASA Systems Engineering principles to develop the Science Traceability and Verification & Validation matrices, ensuring subsystem compliance with mission-level requirements

The Pennsylvania State University

State College, PA

Teaching Assistant – Chem 110

Jan 2023 – May 2023

- Facilitated instruction for CHEM 110, a 300+ student General Chemistry course, delivering targeted concept clarification and strengthening students' foundational understanding
- Led weekly recitation sessions for 40+ students, driving higher problem-solving proficiency and fostering an inclusive, academically engaged learning environment
- Conducted exam review sessions attended by 20–30 students and contributed insights during weekly instructional meetings to align recitation content with course objectives better

Bharat Earth Movers Limited (BEML)

Mysore, IND

Mechanical Engineering Intern

Jul 2023 – Aug 2023

- Researched and developed an advanced lever system for BEML's excavators, reducing operator effort from 10.9 lbs. to 1.7 lbs. through ergonomic redesign and load-path optimization, contributing to a patented innovation
- Conducted R&D on the Roll-Over Protective System (ROPS) for the BG605 Grader, enhancing rollover safety margins by 20% through finite-element validation and field-tested frame reinforcement

LEADERSHIP & ENGINEERING INVOLVEMENT

Penn State Hyperloop Development Team

State College, PA

President & Co-Founder

Aug 2024 – May 2026

- Co-founded and currently lead a 30+ member, 8-discipline team to design and prototype a next-gen hyperloop pod, spearheading sponsorship to secure \$20k by 2026

- Directed the development of a 12-slot dual-sided linear induction motor (DSLIM) and in-house battery management system (BMS), theoretically achieving >68% thrust efficiency

**American Society of Mechanical Engineers (ASME) –
PSU Student Chapter**

Steering & Suspension Team Lead

State College, PA

Jan 2024 – Dec 2024

- Co-led the design and fabrication of TARMO5 RC vehicle suspension and steering systems, achieving a 20% increase in cornering stability and a reduction in body roll through linkage geometry optimization
- Conducted design reviews using ANSYS and SolidWorks Simulation to analyze stress distributions and validate component durability under dynamic loads

SELECTED TECHNICAL PROJECTS

Biomechanical Tremor & Handwriting Analysis Pen – *Medical Mechatronics & Human Systems Design*

Engineered a MEMS-based smart pen integrating accelerometers and force sensors to quantify 4–12 Hz tremors and writing kinematics. Developed a 30-feature ML pipeline achieving >92% stress classification accuracy, bridging biomechanics and embedded system design.

Endotracheal Cuff Pressure Manometer – *Biomedical Instrumentation*

Designed a digital manometer using an ESP32 microcontroller and piezoresistive sensors to monitor 0–40 kPa cuff pressures with $\leq \pm 1\%$ full-scale accuracy. Integrated OLED interface and wireless data logging for real-time surgical feedback.

Wearable Hydration Monitoring System – *Biomechanics & Sensor Design*

Developed a sweat-based wearable integrating microfluidic channels and electrochemical ion sensors for continuous electrolyte tracking. Validated sensor response with artificial sweat samples and achieved <5 mM average deviation in ion concentration readings under dynamic flow conditions.

ENT Surgical Articulating Arm Mount – *Medical Device Design & Ergonomics*

Created a dual-function articulating arm mount for stabilizing endoscopes and surgical microscopes, enhancing surgeon comfort and precision during ENT procedures. Conducted static load tests and mock OR evaluations with clinician feedback to refine locking mechanism stiffness and ease of sterile handling.

High-Performance Shifter Kart – *Design & Manufacturing Engineering*

Designed and fabricated a 205cc shifter kart chassis, increasing simulated torsional rigidity by 15% and acceleration by 25%. Applied CAD/FEA for optimization and validated results via dynamometer and field testing.

TECHNICAL SKILLS

CAD & Design: SOLIDWORKS, Siemens NX, CATIA, Creo, Fusion360, AutoCAD, GD&T

Analysis & Simulation: ANSYS, FEA, Vibrational Analysis, MATLAB/Simulink, Python, NIRfast, K-wave

Robotics/ML/Controls: ROS 2, Vicon Motion Capture, Autonomous Systems, Linux, MATLAB DL Toolbox, Jupyter NB

Manufacturing & Electronics: CNC Lathe/Milling (Sinumerik), Mastercam, Nanofabrication, Arc Welding, FDM (Prusa), MEMS Integration, ESP32, KiCad, Multisim, C/C++